



Building Digital Twins from Interviews: Do AI Interviewers Match Human Fieldwork?

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The Research Problem: Can Digital Twins Match Humans?

LLMs can generate synthetic respondents from **demographic vectors** (Argyle et al., 2023). Feeding **full interview transcripts** improves fidelity (Park et al., 2024). We still do not know how much **interview quality** and **interviewer modality** (human vs. AI) matter for downstream twin accuracy, especially for groups that are under-represented in training data.

Open Question: Are AI-led voice interviews able to collect narratives that are “good enough” relative to in-person human interviews to power accurate digital twins?

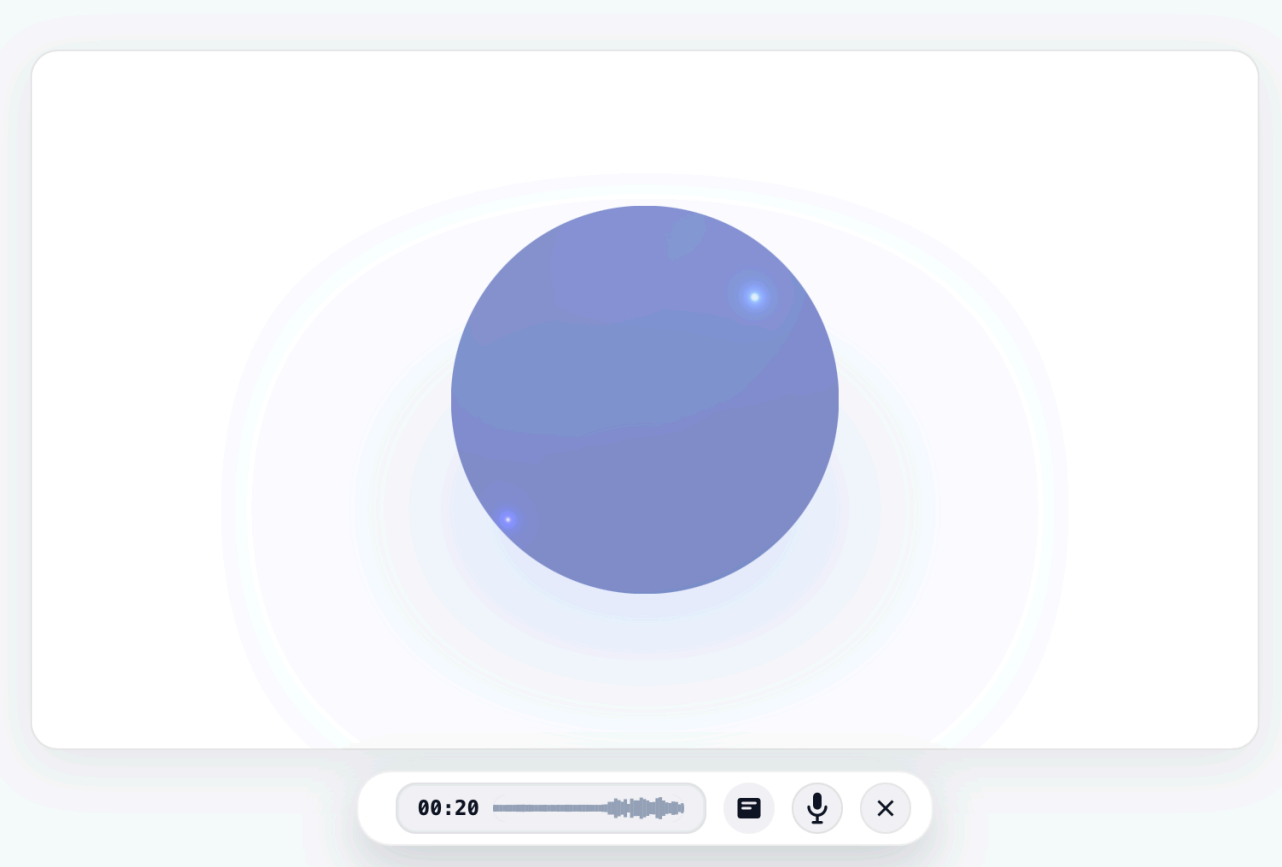
Our Contribution: (1) A head-to-head field test (human vs. AI) using the same semi-structured guide and modules; (2) a low-latency, modular voice chatbot that balances deep probing with smooth topic shifts. We relate interview quality to twin accuracy on held-out items (including second-round vote choice).

Study Design: Human vs. AI Interviewer

We will run a dual-mode study in Santiago, Chile to isolate the **Interviewer Effect** (Human vs. AI) on downstream simulation fidelity.

Two Parallel Samples (N ≈ 700)

- **Condition A (Human):** $N = 200$ in-person interviews conducted by IMFD sociologists.
- **Condition B (AI):** $N = 500$ online voice interviews conducted by “Valentina” (our AI agent) via Netquest.
- **Sampling:** Both arms matched on quotas for Gender, Age, Education, and Commune (Santiago).
- **Instrument:** Shared semi-structured guide with the same follow-up ladders.
 - Human interviewers are trained on it.
 - The AI interviewer mirrors it as closely as possible.
 - Topics: onboarding/rapport; vida/territorio; economía; seguridad; migración; política/valores; elección; socialización; wrap.
- Fielded before the Dec 14, 2025 second-round election to test vote-choice prediction.



Audio-first chat interface.

Platform & Interview Method (Audio, Modules, Control)

We built a voice-first interviewer that feels like a phone call, runs inside Qualtrics, and keeps latency low with the OpenAI Realtime API.

- **Phone-call feel:** low delay; works on laptops or phones with a mic.
- **Topic blocks with a gate:** each block has a goal; the system only advances once it is covered.
- **Auditor:** per-module checklist + min-turn rules; regex move-on cues; lightweight LLM confirms goal met before “ok to advance.”
- **One question at a time:** short asks, natural acknowledgements, quick recap before moving on.

Recipe: How Each Module Runs

1. Present the topic; set expectations.
2. Ask the 2–3 core questions for that module.
3. Loop: listen → if (user says “pasemos/siguiente” **or** auditor says goal met) & min-turns reached, advance; else ask one short follow-up.
4. Recap what you heard in that module.
5. Move to the next module; after the last one, thank and end session.

From Narrative to Simulation (The Digital Twin)

We treat a digital twin as: *an LLM conditioned on one person’s interview that answers new questions “as if” it were that person.*

1. We interview a person (human or AI interviewer) and transcribe it.
2. We feed that transcript to a fresh LLM and ask new questions the person never saw (the held-out battery).
3. The LLM answers “as if” it were that person — that answer set is the digital twin.
4. We compare the twin’s answers to the real person’s answers on the same held-out items to see how faithful the twin is.

What can go into the twin’s context?

- Demographics only (Argyle et al. baseline)
- Demographics + raw transcript (Park et al. style baseline)
- Optional add-ons: brief module recaps (1–2 lines), 1–3 key quotes, or a short summary if the transcript is long
- Held-out battery prompts (the questions the twin must answer)

Evaluation: Defining “Quality”

We score two dimensions: *Process Quality* (the interview) and *Outcome Fidelity* (the twin).

A. Interview Quality

- **Coverage:** Share of checklist items elicited.
- **Depth:** How detailed the answers are.
- **Flow:** Stays on topic; keeps to one question at a time (avoids stacking).

B. Simulation Fidelity

- **Accuracy:** Match rate on held-out items (incl. runoff vote choice).
- **Consistency:** Answers to related trade-offs line up with each other.

Hypotheses & Analytical Plan

- **H1 – Depth Gap:** AI-led interviews capture less coverage/depth than human-led ones.
- **H2 – Fidelity Gap via Depth:** Twins built from AI transcripts are less accurate, and most of that gap is explained by coverage/depth.
- **H3 – Cost-Fidelity Trade-off:** AI interviews deliver “good enough” twin accuracy (e.g., ~85–90% of human) while using far fewer minutes/cost.

Planned tests

- Check how many checklist items and how much detail each mode gets (and how many turns it takes).
- See whether differences in coverage/depth explain the gap in twin accuracy.
- Judge whether the AI twin is “good enough” once we factor in the lower time and cost.

Status & Next Steps

- **Current:** Recruiting human interviewers and live-testing the AI interviewer to validate coverage, flow, and auditor gating.
- **Fieldwork goal:** Deploy both modes before the Dec 14, 2025 runoff to enable vote-choice prediction; results targeted March 2026.

Ethics & Privacy

- Consent covers using transcripts to build the simulated agent; participation is voluntary and questions may be skipped.
- Transcripts are de-identified; audio stored separately with limited retention; withdrawal honored.
- No fine-tuning on respondent data.

References

Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., & Wingate, D. (2023). *Out of One, Many: Using Language Models to Simulate Human Samples*. *Political Analysis*, 31(3), 337–351. <https://doi.org/10.1017/pan.2023.2>.
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